



NUCLEAR POWER CORPORATION OF INDIA LIMITED

WELDING PROCEDURE SPECIFICATION (WPS) - ASME QW-482

SECTION 1 - IDENTIFICATION			
WPS NO.	NPCIL-GTSM-11H	REVISION NO.	R-1
DATE	2014-07-07	QUALIFICATION STATUS	To be qualified
SUPPORTING PQR NO.(S)	To be made by user organization		

SECTION 2 - WELDING PROCESSES			
WELDING PROCESS (ROOT)	GTAW	WELDING PROCESS (FILL)	GTAW+SMAW
TYPE	Manual	JOINT DESIGN	Groove
BACKING	With or Without	BACKING MATERIAL	As per design document

SECTION 4 - BASE METALS (QW-403)			
P-NO.	P-No. 1 Group 1/2	TO P-NO.	P-No. 1 Group 1/2
MATERIAL SPEC / GRADE	Synthetic but code-style material family		
THICKNESS RANGE (GROOVE)	19 mm to 38 mm	THICKNESS RANGE (FILLET)	All Thickness

SECTION 5 - POSITIONS (QW-405)			
POSITION(S) OF GROOVE	All	WELDING PROGRESSION	Vertical Uphill
POSITION(S) OF FILLET	All	PWHT	Required

PREPARED BY

REVIEWED BY (QA)

APPROVED BY



SECTION 6 - PREHEAT & PWHT (QW-406 / QW-407)

PREHEAT TEMP, MIN	121.0 °C	INTERPASS TEMP, MAX	250.0 °C
PWHT REQUIREMENTS	Temperature Range: 595-675 °C		

SECTION 8 - ELECTRICAL CHARACTERISTICS (QW-409)

CURRENT / POLARITY	DC-EN
TRAVEL SPEED RANGE	40-80 mm/min

SECTION 9 - TECHNIQUE (QW-410)

STRING OR WEAVE BEAD	Root: Stringer / Fill: Weave
CLEANING METHOD	Brushing or Grinding
PEENING ALLOWED	False

SECTION 10 - FILLER METALS (QW-404)

SFA SPECIFICATION	5.18+5.1	AWS CLASSIFICATION	ER70S-2 + E7018
F-NO.	varies	A-NO.	varies

NOTES & RESTRICTIONS:

Synthetic WPS record used for contrast and edge cases.

JOINT SCHEMATIC

